

Two Dimensional Plots

1 Plotting Functions and Equations

- the graph of a function
- equations in two variables

2 Curves in Parameter Form

- rectangular coordinates
- polar coordinates

MCS 320 Lecture 26
Introduction to Symbolic Computation
Jan Verschelde, 17 July 2026

Two Dimensional Plots

1 Plotting Functions and Equations

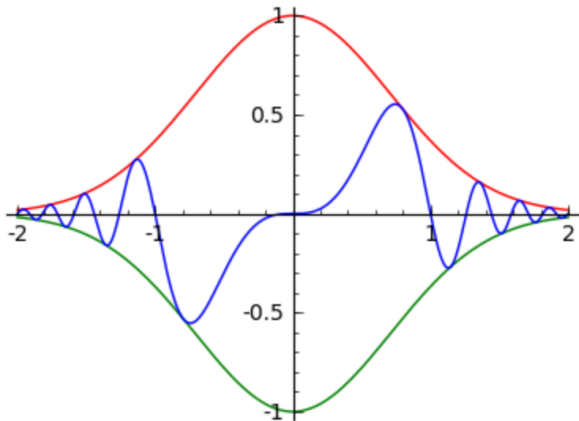
- the graph of a function
- equations in two variables

2 Curves in Parameter Form

- rectangular coordinates
- polar coordinates

the Graph of a Function $y = f(x)$

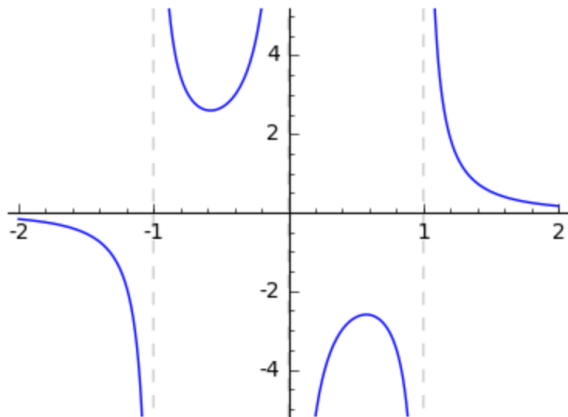
The function $f(x) = \exp(-x^2) \sin(\pi x^3)$ appears in blue below:



The plots of $\pm \exp(-x^2)$ show the decaying amplitude of $f(x)$.

Poles and Vertical Asymptotes

Plotting $f(x) = 1/(x^3 - x)$ requires some special care ...



Two Dimensional Plots

1 Plotting Functions and Equations

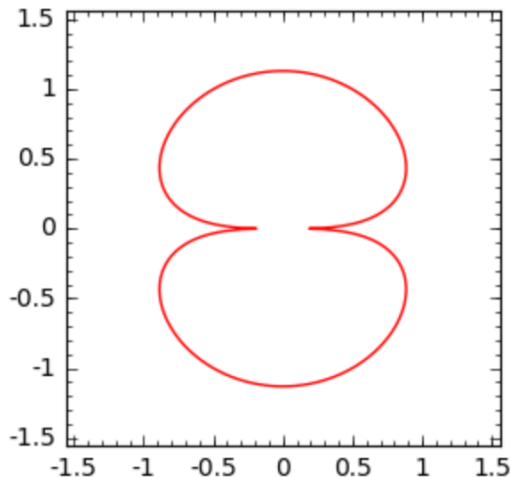
- the graph of a function
- equations in two variables

2 Curves in Parameter Form

- rectangular coordinates
- polar coordinates

A Curve Invented by James Watt

$$f(x, y) = (x^2 + y^2)^3 + 5.12(x^2 + y^2)^2 - 5.15(x^4 - y^4) - 14.7456y^2 = 0$$



Two Dimensional Plots

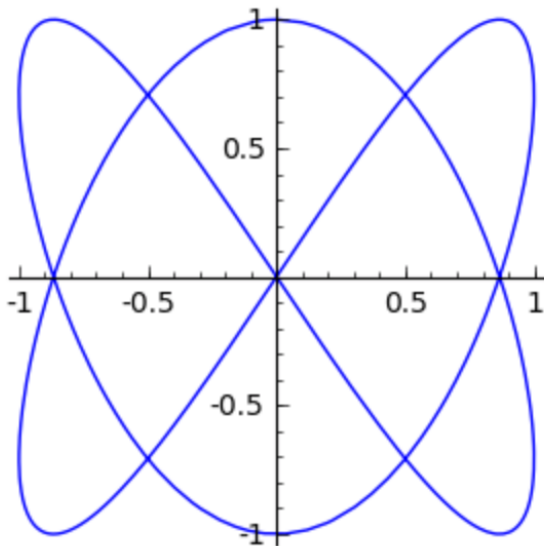
1 Plotting Functions and Equations

- the graph of a function
- equations in two variables

2 Curves in Parameter Form

- rectangular coordinates
- polar coordinates

A Lissajous Curve ($x(t) = \sin(2t)$, $y(t) = \sin(3t)$)



Two Dimensional Plots

1 Plotting Functions and Equations

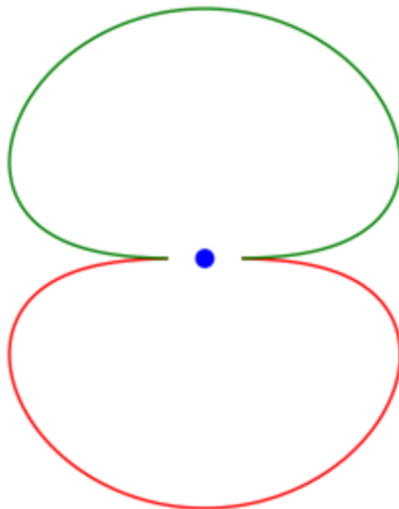
- the graph of a function
- equations in two variables

2 Curves in Parameter Form

- rectangular coordinates
- **polar coordinates**

Polar Coordinates $r = f(t)$, $t \in [0, 2\pi]$

The curve invented by James Watt in polar coordinates:



from rectangular to polar coordinates

Given is an equation $f(x, y) = 0$.

A polar representation can be obtained as follows:

- 1 For radius r and angle t , compute

$$g(r, t) = f(x = r \cos(t), y = r \sin(t)).$$

- 2 Simplify g using trigonometric identities.

- 3 Solve $g(r, t) = 0$ for r .

The solutions of $g(r, t) = 0$ define the components of the curve.

If the curve passes several times through the origin,
then the polar representation may greatly simplify the plotting.